

BUPT Compilers Lab1 2024Fall

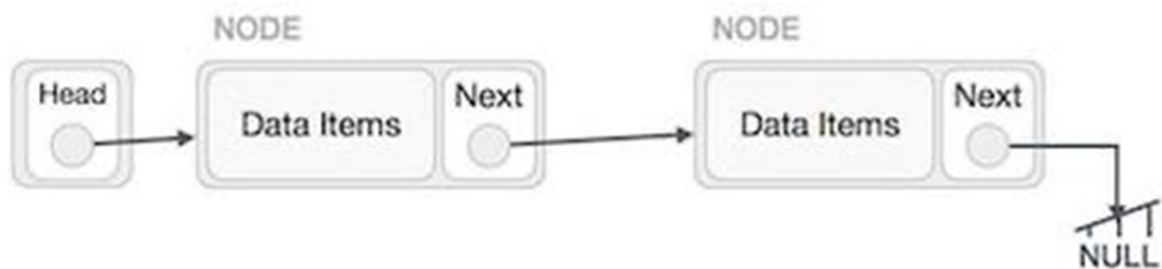
Overview

Welcome to the lab class! In this course, you will be completing the programming assignments. Lab1 is a C language exercise with the objective of implementing a linked list structure using the C programming language. This lab serves as a preparatory exercise and will help you familiarize yourself with pointer operations and memory management. In the context of compiler development, linked lists are instrumental in efficiently organizing and managing data structures such as symbol tables and syntax trees, which is crucial for subsequent compiler experiments.

Linked List

For beginners:

C-language interactive tutorial: <https://www.learn-c.org/>



- Each node contains two fields.
- For header node, the first field contains the number of nodes in the linked list, excluding itself.
- For other nodes, the first field contains an int value.
- The second field is a pointer, points to the next node.

Coding

1. You will complete the following functions (declared in header file) in the linked_list.c file!

```

/* insert val at position index */
void linked_list_insert(node *head, int val, int index);

/* delete node at position index */
void linked_list_delete(node *head, int index);

/* remove the first occurrence node of val */
void linked_list_remove(node *head, int val);

/* remove all occurrences of val */
void linked_list_remove_all(node *head, int val);

/* get value at position index */
int linked_list_get(node *head, int index);

/* search the first index of val */
int linked_list_search(node *head, int val);

/* search all indexes of val */
node *linked_list_search_all(node *head, int val);

```

2. Make the libll target

```

> /mnt/g/4xDeskTop/BUPT/CompilerPrinciple/homework/lab1/lab1 > ...
make libll
gcc linked_list.c --shared -fPIC -o libll.so

> /mnt/g/4xDeskTop/BUPT/CompilerPrinciple/homework/lab1/lab1 > ...

```

3. Run self-test examples in python3

Most tests failed, try to pass them!

```

=====
FAIL: test_search (__main__.LinkedListTest)
-----
Traceback (most recent call last):
  File "ll_test.py", line 141, in test_search
    self.assertTrue(idx == i)
AssertionError: False is not true

=====
FAIL: test_search_all (__main__.LinkedListTest)
-----
Traceback (most recent call last):
  File "ll_test.py", line 151, in test_search_all
    self.assertEqual(str(ret), '7->12')
AssertionError: ' ' != '7->12'
+ 7->12

-----
Ran 8 tests in 0.003s

FAILED (failures=7)

```

The result is expected to be like this.

```

C:\mnt\g\4xDeskTop\BUPT\编译原理\作业\lab1\lab1 > python3 ll_test.py
.....
Ran 8 tests in 0.002s

OK

D:\mingw-w64\bin\gcc.exe --shared -TPIC -o lllbc.dll linked_list.c
admin@4XWi11 G: > 4xDeskTop > BUPT > 编译原理 > 作业 > lab1 > lab1
> python3 .\ll_test.py
.....
Ran 8 tests in 0.003s

OK
admin@4XWi11 G: > 4xDeskTop > BUPT > 编译原理 > 作业 > lab1 > lab1
>

root@599e0e665b42:/home/labs/lab1# python3.6 ll_test.py
.....
Ran 8 tests in 0.001s

OK
root@599e0e665b42:/home/labs/lab1#

```

Report

Additionally, you will need to submit a pdf report(name_studentID.pdf) documenting your work for assignment. Please carefully follow the instructions outlined below:

1. Academic Integrity: Plagiarism or any form of cheating is strictly prohibited. Your work should be original, and any external sources should be appropriately cited.
2. Programming Assignments: Feel free to ask questions and seek assistance from the teaching assistant if needed.
3. Report:
 - Pdf type
 - Naming like **name_studentID.pdf**
 - Include any relevant diagrams, charts, or screenshots to enhance your explanations.

- Make sure your report is well-structured, with appropriate headings and subheadings.

4. Submission Guidelines:

- Include your **name, student ID, class number and container number** in the report's header. For Docker on Windows systems, you can view the container numbers in the containers of the Docker Desktop.
- Commit the compressed package of the lab1 folder (**lab1.zip**), which should include the code you wrote, the compilation results, and your PDF report.

5. Deadline:

September 15, 2024, 24:00

6. Submission Platform:

Teaching cloud platform

Report format

Name: xxx

Student ID: xxxxxxxxxx

Class Number: xxxxxxxxxx

Container Number: 04c56e0ced1b3944f2eeee027840f5210f54efecc49b3377166c9a7a32ff119